

Analog & Digital Communication (EE3003)

Date: September 23, 2024

Course Instructor(s)

1. Dr. S. M. Sajid (CM)

2. Mohsin Yousuf

Sessional-I Exam

Total Time (Hrs): 1

Total Marks: 40

Total Questions: 3

Roll No

Section

Stud

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1. Attempt all questions and remember to solve parts of the same question together.
2. Show all the steps with the help of diagrams and equations.

CLO # 01: Recognize the parameters of a communication system and the noise/distortion therein.

Q1:

[10 marks]

(a) State Shannon's theorem and clearly **define** all the terms used therein.

[5]

(b) Provide the relationship between SNR (Signal-to-Noise ratio) and B (Bandwidth) in the simplest terms.

[2]

(c) If SNR is increased by a factor of 4 in part (b), **compute** the effect on the corresponding bandwidth.

[3]

CLO # 01: Recognize the parameters of a communication system and the noise/distortion therein.

Q2: Signals $g_1(t) = 5000 \text{ rect}(5 \times 10^3 t)$ and $g_2(t) = \delta(t)$ are applied at the inputs of ideal low-pass filters (LPFs); $H_1(\omega) = \text{rect}(\omega/20,000\pi)$ and $H_2(\omega) = \text{rect}(\omega/10,000\pi)$. The outputs of $y_1(t)$ and $y_2(t)$ of these filters multiplied to obtain the signal $y(t) = y_1(t)y_2(t)$. Answer the following:

[15 marks]

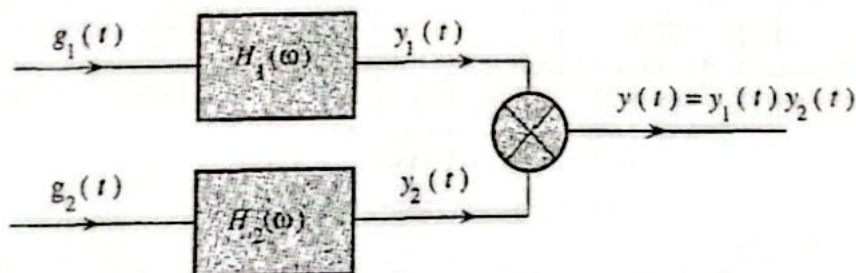


Figure 1: Signals passing through LPF and multiplier

- (a) Sketch $G_1(\omega)$ and $G_2(\omega)$. [4]
 (b) Sketch $H_1(\omega)$ and $H_2(\omega)$. [4]
 (c) Sketch $Y_1(\omega)$ and $Y_2(\omega)$. [4]
 (d) Identify $Y(\omega)$ in terms of $Y_1(\omega)$ and $Y_2(\omega)$ and comment on its bandwidth. [3]

CLO # 02: Compute energy and power spectral density of the signal.

Q3:

[15 marks]

- (a) For the signal,

[8]

$$g(t) = \frac{2a}{t^2 + a^2}$$

Determine the essential bandwidth B Hz of $g(t)$ such that the energy contained in the spectral components of $g(t)$ of frequencies below B Hz is 97% of the signal energy.

Hint:

$$e^{-a|t|} \Leftrightarrow \frac{2a}{a^2 + \omega^2}$$

Use symmetry property: $G(t) \Leftrightarrow 2\pi g(-\omega)$ to find the spectrum $G(\omega)$ and ESD $\Psi_g(\omega)$ of the given signal.

$$E_g = \frac{1}{\pi} \int_0^{\infty} |G(\omega)|^2 d\omega$$

- (10) Figure 2 shows a random binary pulse train $g(t)$. The pulse width is $T_b/2$ and the amplitude is A . One binary digit is transmitted every T_b seconds. A binary 1 is transmitted by the positive pulse A , and a binary 0 is transmitted by the negative pulse $-A$. The two symbols are equally likely and occur randomly.

[7]

Determine the autocorrelation function and PSD.

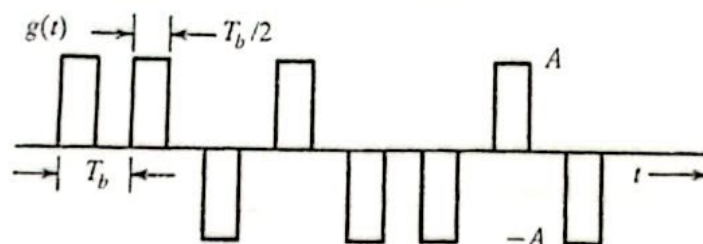


Figure 2 A binary pulse train